

spontaneously terminates actuating motion. Aiming as a self-regulated actuation is modulated by the negative feedback loop inherent in tight hydrogel-stimuli interactions. Supporting an arrangement of solar cells with SunBOT stem is shown in Fig. 6.

Smart hydrogels

Polymers are long chains of chemically-linked repeat units called monomers. Hydrogels are binary polymers, and are capable of absorbing and retaining water up to twenty times their dry weight. These are flexible, too.

Hydrogels can be conventional or stimulus-responsive. Stimulus-responsive hydrogels are called smart hydrogels and rapidly change their equilibrium state in response to external stimuli such as pH, temperature, electric field, magnetic field, light, pressure and so on.

Smart hydrogels responsive to heat are called thermo-responsive hydrogels, and solubility of polymer changes with temperature. Lower and upper boundaries of solubility are

called LCST and upper critical solution temperature (UCST), respectively.

Plasmon resonance

Nanoparticles of noble metal, graphene, etc, whose sizes are comparable to incident light wavelength, interact with light by Mie scattering. Mie scattering suggests situations where size of the scattering particles is comparable to wavelength of the light, rather than much smaller or much larger.

Mie scattering arises when an electromagnetic wave incident on the particles and all ranges of wavelengths get reflected back equally to the atmosphere. Collective oscillations of free surface electrons are called plasmon. Nanoparticles of noble metal and graphene support surface plasmons, that is, oscillations of the conduction electrons at the nanoparticle surface that result in extraordinary optical properties that are not exhibited by any other class of material.

Light absorbed by the particles excites the free electrons. Specific wavelengths of light can drive conduction electrons in the nanoparticles having size around resonance frequency of light, to collectively oscillate—a phenomenon known as surface plasmon resonance (SPR). When these resonances are excited, absorption and scattering intensities can be up to forty times higher than identically-sized other nanoparticles that are not plasmonic.

The photo-responsive nanoparticles of gold/graphene absorb light

and generate heat through plasmon resonance, which is transferred to the surrounding hydrogel. When heat generated raises the temperature more than LCST of the smart hydrogel, the hydrogel starts to shrink. **Bending of SunBOT stem is achieved by differential shrinking across the cross-section. The part closest to the light shrinks the most, while the part away from the light shrinks the least.**

Tracking performance

Stimulated thermal and stretch fields within SunBOT pillars reveal that this wide-angle tracking ability is induced primarily by contraction at the illuminated region rather than governed by the material stiffness or flexural modulus, as in a typical elastic beam bending. This allows the freedom in material selection.

Further, phototropism of SunBOT is energy-efficient. As this light-powered hydrogel configuration is essentially a volumetric phase transition with a tunable LCST, it consumes a minute amount of photonic power to locally heat the hydrogel and trigger the release of a large amount of mechanical energy for phase transition.

SunBOT has a truly broad and tunable operation window that can guarantee successful phototropism over a wide range of light intensities, ambient temperatures, lighting conditions, irregular geometric shapes and rhythms of illumination. These can be achieved by tuning material properties, nano-photo-absorber concentration, transition temperature, dimensions of the material, etc. It can not only track a spotlight but also large area lighting.

The hydrogel has a fast recovery when not being illuminated. It can successfully track a light source in real time, continuously moving 360 degrees azimuthally at a fast speed up to 0.03 second per degree, and can precisely stop at the light source direction without over- or under-bending. An array of SunBOTS can, in principle, achieve up to 400 per cent solar energy, harvesting enhancement over non-tropic materials at oblique illumination angles.

Other concepts of heliotropic solar cell

There are other variants of passive heliotropic and phototropic systems

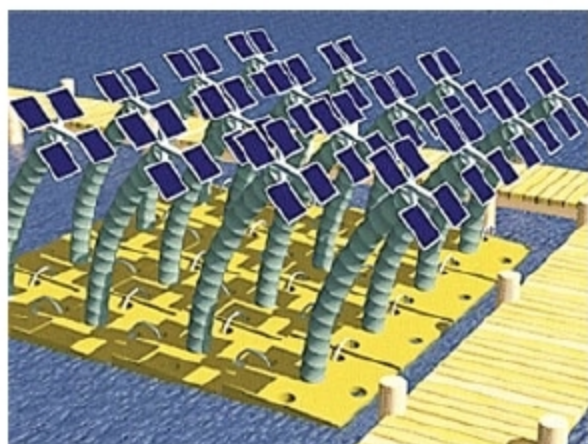


Fig. 6: Fixing arrangement of solar panel with SunBOT stem

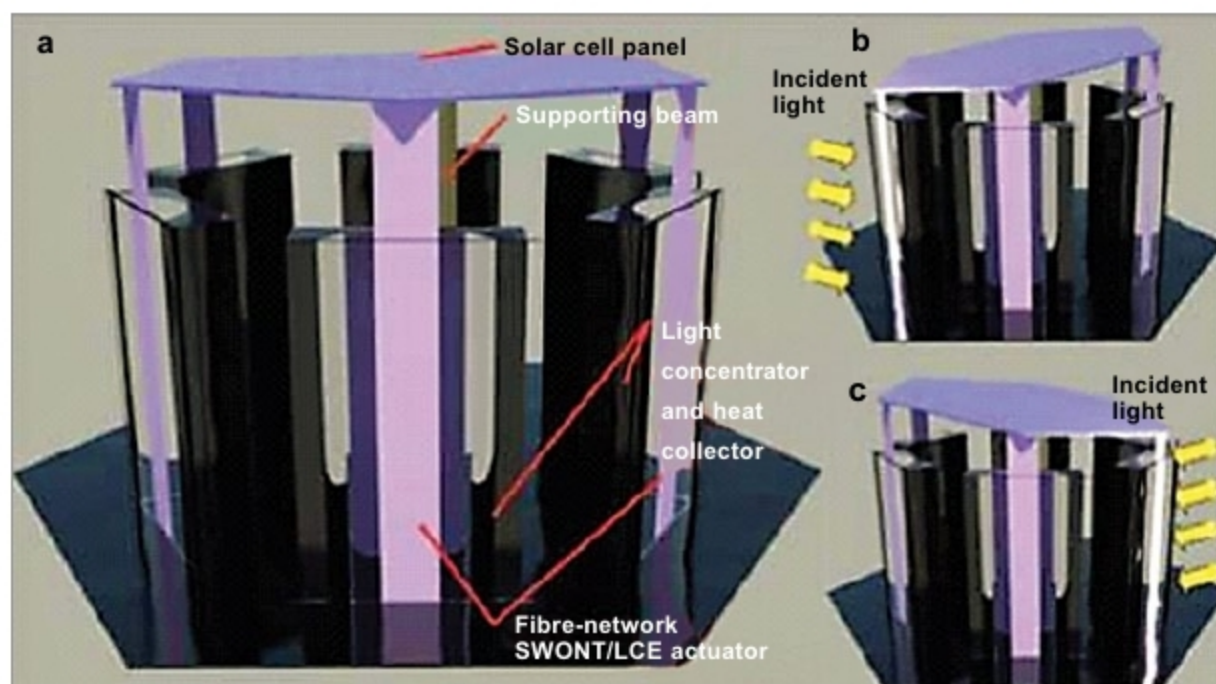


Fig. 7: Alternative arrangement of heliotropic solar panel supported by composite beam